

**OMNICHANNEL RETAIL SALES AND INVENTORY
ANALYTICS DASHBOARD
FINAL PROJECT REPORT**

1. Problem Statement :

Retail businesses operating across both physical stores and online platforms face challenges due to fragmented data, making it difficult to track sales performance, manage inventory efficiently, and identify key trends. This project aims to develop a unified data analytics pipeline and interactive dashboard that consolidates multi-channel retail data, enabling real-time insights into sales trends, customer behavior, and inventory performance to support data-driven decision-making.

2. Tools Used :

- Power BI - Dashboard creation & visualization
- SQL - Data extraction & analysis
- Excel - preprocessing
- Python – Data Cleaning

3. Dataset Description :

The dataset contains sales transaction data including:

- Transaction_ID
- Date
- Customer_Name
- Product
- Total_Items
- Total Cost
- Payment Method
- City
- Store Type
- Discount Applied
- Customer Category Season
- Promotion

4. Data Cleaning Steps :

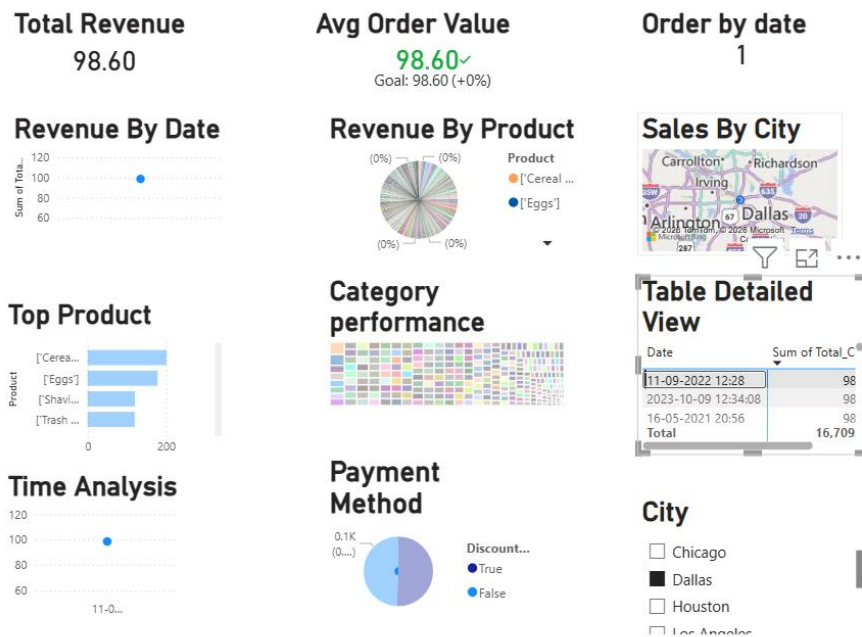
- Removed null and duplicate values
- Converted date column into proper format
- Created new columns: Year, Month, Day
- Checked data consistency (price, quantity)
- Ensured correct data types for analysis

5. SQL Queries:

- **created database**
CREATE DATABASE retail_analytics;
- **Use Retail Analytics Dataset**
use retail_analytics;
- Revenue of Data
SELECT SUM(Total_Cost) AS total_sales
FROM retail_transactions_dataset;
- Revenue by city
SELECT city,
SUM(Total_Cost) AS total_revenue
FROM retail_transactions_dataset

GROUP BY city
ORDER BY total_revenue DESC;

6. Dashboard Screenshot :



Insight

- "This dashboard provides a unified view of omnichannel retail performance by combining sales, customer, and time-based insights.
- It reveals that revenue is highly seasonal, driven by top-performing products and major cities with

7. Business Insight :

- **Identify peak revenue periods** (monthly/weekly/daily)
→ Example: Sales may spike during **Q4 (festive season)**
- **Top-performing days**
→ Weekends or specific weekdays (e.g., Friday/Saturday) may generate higher revenue
- **Average Order Value (AOV) trends**
→ Helps understand customer spending behavior
Insight: Increase marketing during high-performing periods to maximize revenue
- High demand, fast-moving inventory
- Promote top products more
- Discount or bundle slow-moving items to clear inventory

8. Recommendation :

- Optimize inventory
- Run targeted promotions
- Focus on high-performing regions

9. Conclusion :

The dashboard provides a comprehensive view of sales performance and helps businesses make data-driven decisions.

It improves visibility into trends, customer behaviour, and operational efficiency.

